

Vinh Ly

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Summary

Software Engineer with **18 years** building distributed systems at scale. Led video infrastructure serving billions of daily queries at YouTube and search ranking systems at Google with p99 latency under 200ms. Reduced infrastructure costs by \$12M annually through monolith-to-microservices migration. Achieved 99.9% uptime for systems processing 100M+ events daily. Developed evaluation frameworks for autonomous vehicle simulation and partnered with safety, systems, and infrastructure teams. Next focus: evaluation algorithms for autonomous vehicles. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Distributed Systems: Kubernetes, microservices, SLO, on-call, capacity planning, distributed tracing

Evaluation Infrastructure: statistical analysis, simulation, metrics frameworks, regression testing, data pipelines

Languages: C++, Python, Go, Java, C#, TypeScript, JavaScript, HTML, CSS

Data Platforms: PostgreSQL, BigQuery, Cassandra, MapReduce, data warehousing

Work Experience

Staff Software Engineer at YouTube

02/2025 - Present

- Owned video infrastructure serving billions of daily queries with p99 latency under 200ms and 99.9% uptime.
- Led reliability initiatives reducing incidents by 30% through improved monitoring and incident response processes.
- Partnered with video playback and content delivery teams to optimize user experience at scale.
- Designed data pipelines processing 10+TB daily for video analytics and performance reporting.
- Established engineering standards for distributed systems development across multiple teams.

Staff Software Engineer at Whatnot

05/2024 - 02/2025

- Built **Go** services handling 50k+ concurrent connections for live auction platform during rapid growth phase.
- Reduced checkout latency by 45% through **PostgreSQL** query optimization and multi-layer caching implementation.
- Designed real-time inventory tracking system processing 100k+ events per second with 99.9% uptime.
- Implemented auction settlement pipeline supporting \$500M+ in annual transactions with end-to-end consistency guarantees.
- On-call rotation for critical path services maintaining SLO adherence above 99.9%.

Staff Software Engineer/Manager at Google

03/2015 - 04/2024

- Led team of 8 engineers across multiple locations while partnering with product and infrastructure teams.
- Directed migration from monolith to **microservices**, improving deployment frequency from monthly to weekly.
- Reduced infrastructure costs by \$12M annually through **capacity planning** and **resource optimization**.

- Built **distributed systems** serving 3.5B+ daily queries, achieving p99 latency under 200ms.
- Designed high-performance **C++** services processing 100M+ events daily with reliability above 99.9%.

Software Engineer at Google

02/2011 - 02/2015

- Built data pipelines handling 10+TB daily and designed APIs serving 100+ internal teams.
- Maintained 99.9% uptime for core services while mentoring 5 junior engineers.
- Reduced manual analysis time by 60% through automated reporting infrastructure.

Software Design Engineer II at Microsoft

10/2007 - 12/2011

- Developed **C++** Windows kernel drivers and **Win32 API** applications for core OS functionality.
- Built test automation frameworks using **C#**, increasing coverage and reducing regression cycles.
- Collaborated with kernel and security teams to resolve low-level system issues.

Software Design Engineer Intern at Microsoft

06/2006 - 08/2006

- Developed internal tooling and automation scripts to support team operations.

Education

B.S. Honors degrees in Computer Science

Virginia Tech | 2004 - 2007